The ARIMA Procedure

Name of Variable = Pr	
Mean of Working Series	6.9335
Standard Deviation	1.828232
Number of Observations	20

Squared Canonical Correlation Estimates						
Lags	MA 0	MA 1	MA 2	MA 3	MA 4	MA 5
AR 0	0.9267	0.7736	0.5636	0.3429	0.1577	0.0258
AR 1	0.2606	0.1776	0.1489	0.2082	0.2672	0.2321
AR 2	0.0379	0.0015	0.0073	0.0035	0.0063	0.0178
AR 3	0.0138	0.0035	0.0034	0.0009	0.0005	0.0021
AR 4	0.0322	0.0046	0.0010	0.0009	0.0020	0.0001
AR 5	0.0148	0.0081	0.0006	<.0001	0.0006	0.0033

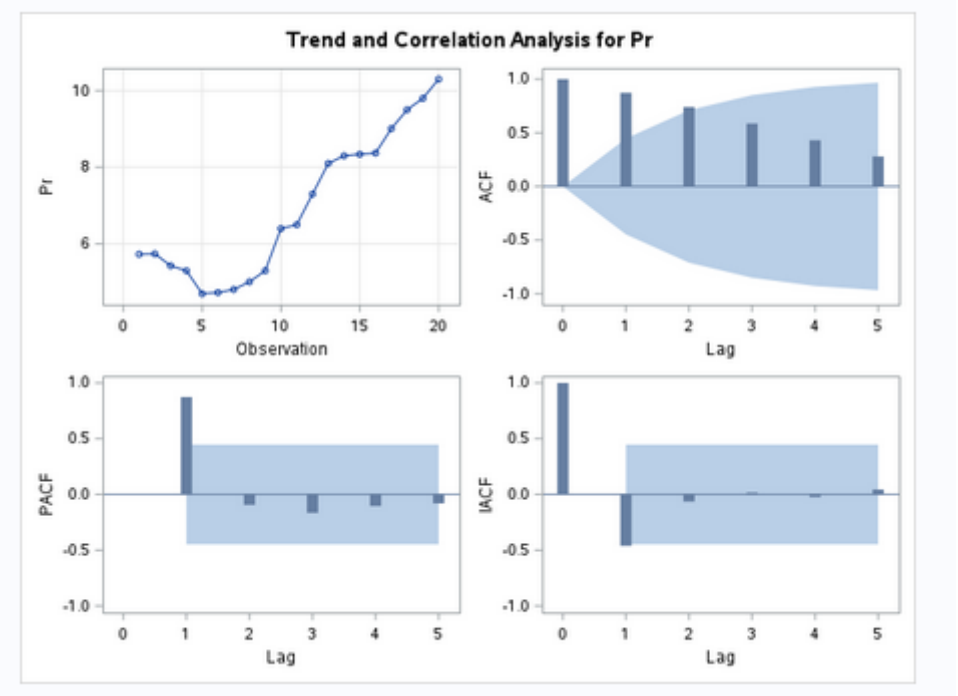
SCAN Chi-Square[1] Probability Values						
Lags	MA 0	MA 1	MA 2	MA 3	MA 4	MA 5
AR 0	<.0001	0.0084	0.0769	0.2132	0.4102	0.7374
AR 1	0.0166	0.0940	0.1925	0.1709	0.2757	0.2773
AR 2	0.4044	0.8951	0.7750	0.8476	0.8097	0.6988
AR 3	0.6271	0.8543	0.8628	0.9255	0.9436	0.8886
AR 4	0.4692	0.8387	0.9231	0.9294	0.9050	0.9745
AR 5	0.6366	0.7941	0.9393	0.9920	0.9476	0.8799

Extended Sample Autocorrelation Function						
Lags	MA 0	MA 1	MA 2	MA 3	MA 4	MA 5
AR 0	0.8735	0.7405	0.5865	0.4304	0.2793	0.1220
AR 1	0.2849	0.2682	0.1702	0.0424	-0.2641	-0.3196
AR 2	0.5149	0.1134	0.1927	-0.0037	-0.3631	-0.0264
AR 3	-0.0341	0.5003	-0.0214	-0.0250	-0.4305	-0.0503
AR 4	0.7227	0.3811	-0.2122	-0.1410	-0.3596	-0.0040
AR 5	0.5607	0.3102	-0.0921	-0.1916	-0.4619	.

ESACF Probability Values						
Lags	MA 0	MA 1	MA 2	MA 3	MA 4	MA 5
AR 0	<.0001	0.0372	0.1682	0.3539	0.5637	0.8041
AR 1	0.2143	0.2912	0.5400	0.8802	0.3206	0.2534
AR 2	0.0289	0.6507	0.5412	0.9893	0.2788	0.9201
AR 3	0.8883	0.0650	0.9469	0.9183	0.1979	0.8688
AR 4	0.0038	0.1339	0.3990	0.6333	0.3454	0.9933
AR 5	0.0299	0.2344	0.7499	0.6051	0.2405	.

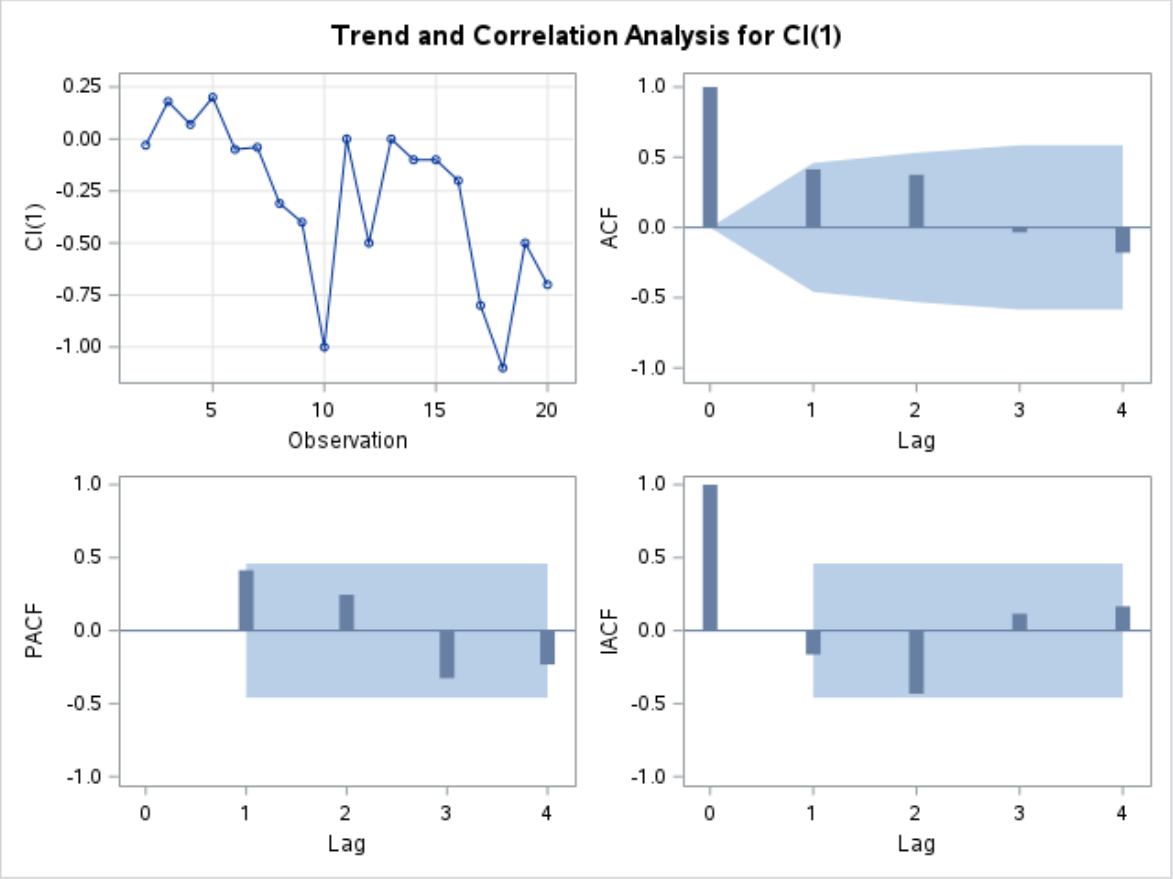
ARMA(p+d,q) Tentative Order Selection Tests			
SCAN		ESACF	
p+d	q	p+d	q
1	1	1	0
2	0	3	0
0	2	0	2
		5	1

(5% Significance Level)



The ARIMA Procedure

Name of Variable = CI	
Period(s) of Differencing	1
Mean of Working Series	-0.28316
Standard Deviation	0.378015
Number of Observations	19
Observation(s) eliminated by differencing	1



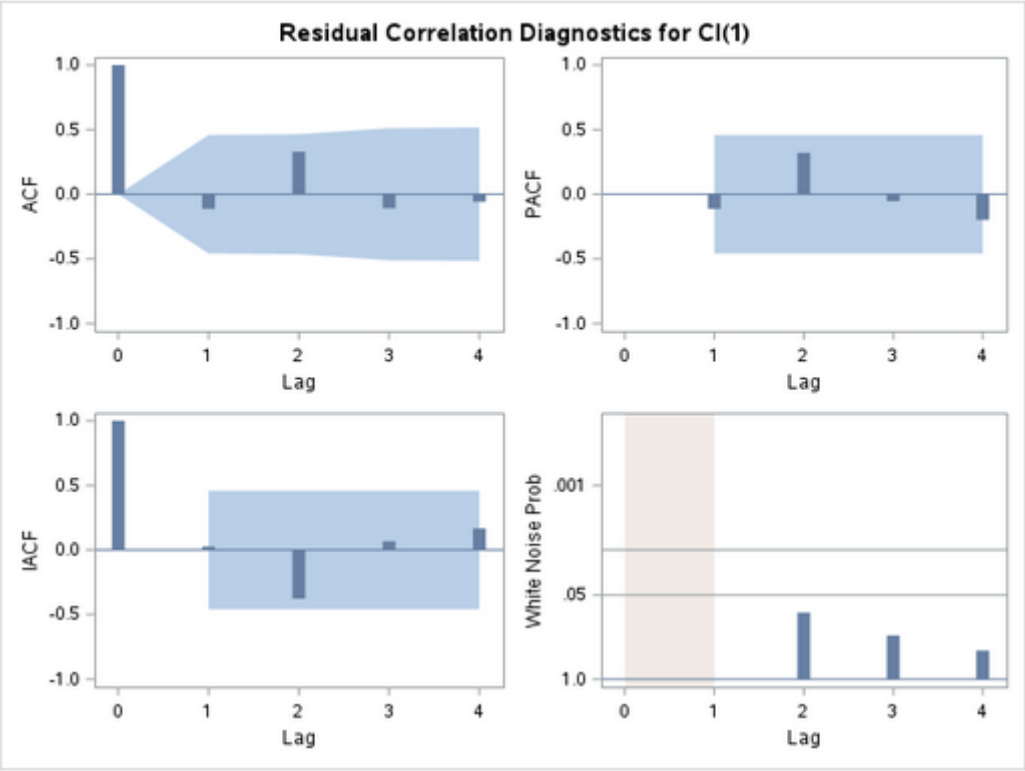
Conditional Least Squares Estimation					
Parameter	Estimate	Standard Error	t Value	Approx Pr > |t|	Lag
MU	-0.28181	0.14070	-2.00	0.0614	0
AR1,1	0.44193	0.22701	1.95	0.0683	1

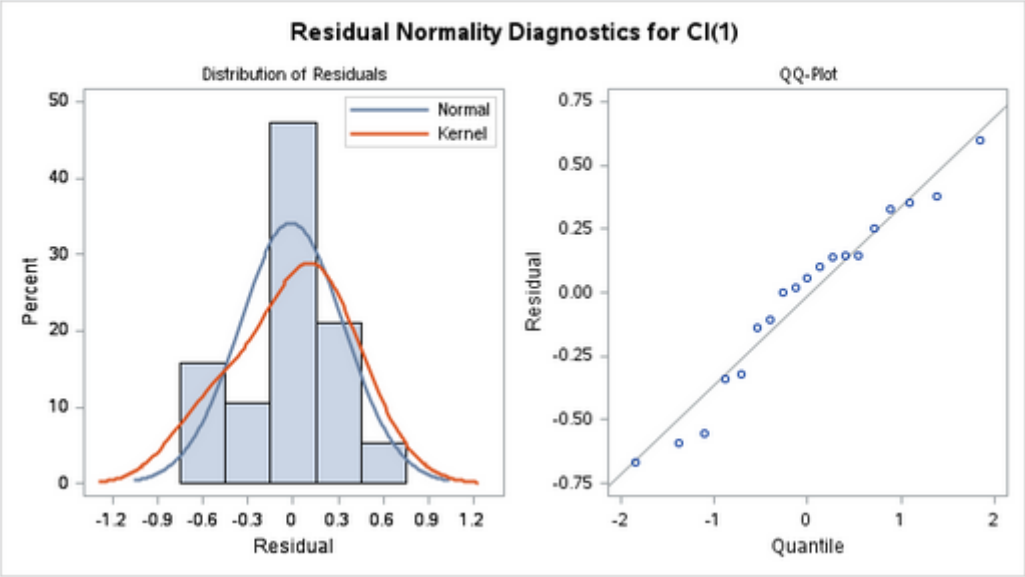
Constant Estimate	-0.15727
Variance Estimate	0.130528
Std Error Estimate	0.361286
AIC	17.11912
SBC	19.008
Number of Residuals	19

* AIC and SBC do not include log determinant.

Correlations of Parameter Estimates		
Parameter	MU	AR1,1
MU	1.000	-0.053
AR1,1	-0.053	1.000

Autocorrelation Check of Residuals									
To Lag	Chi-Square	DF	Pr > ChiSq	Autocorrelations					
6	4.49	5	0.4811	-0.112	0.332	-0.108	-0.059	-0.209	-0.019
12	11.10	11	0.4350	-0.197	0.350	-0.020	0.130	0.024	-0.035
18	18.26	17	0.3726	-0.131	-0.108	-0.194	-0.066	-0.035	-0.037



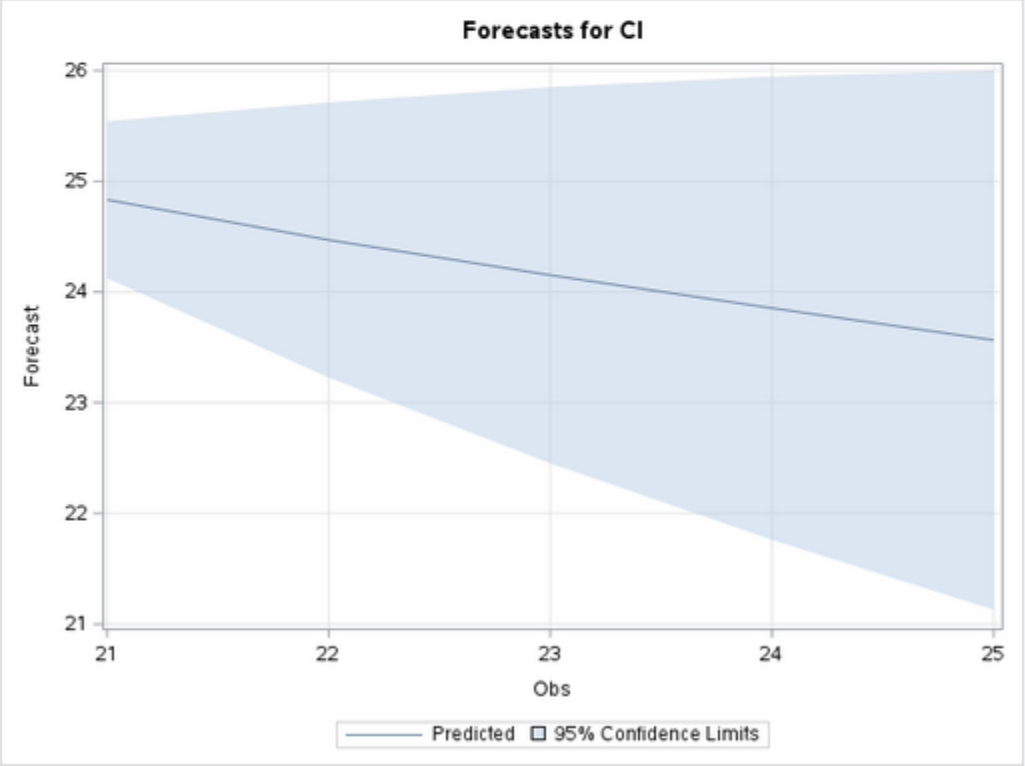


Model for variable CI	
Estimated Mean	-0.26181
Period(s) of Differencing	1

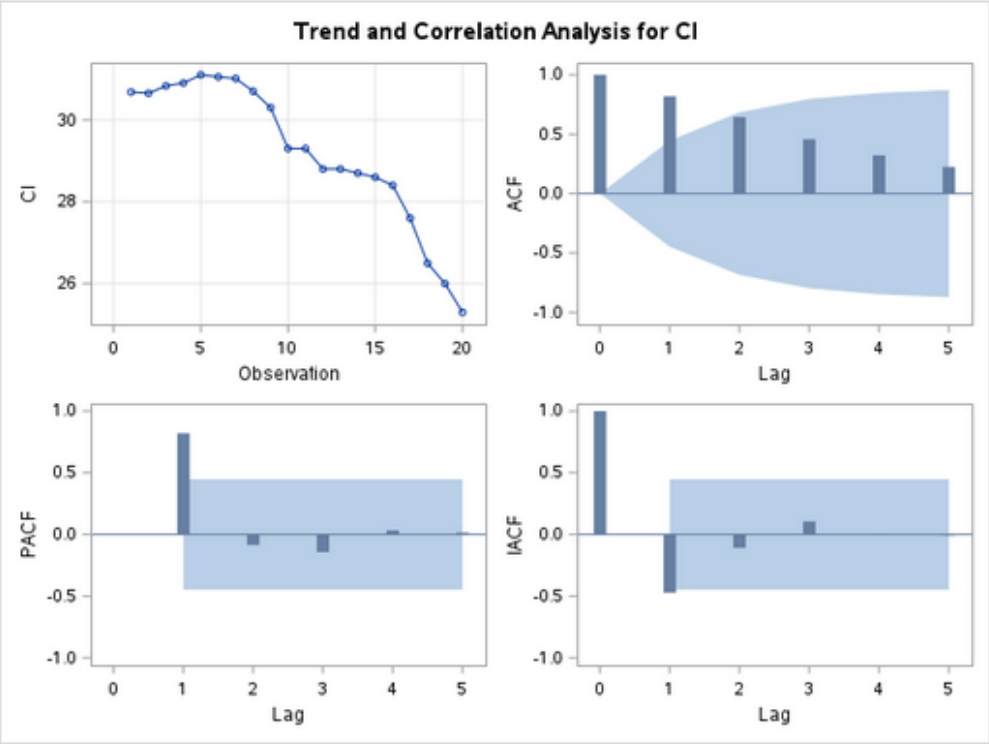
Autoregressive Factors	
Factor 1:	1 - 0.44193 B**(1)

Forecasts for variable CI

Obs	Forecast	Std Error	95% Confidence Limits	
21	24.8334	0.3613	24.1253	25.5415
22	24.4699	0.6340	23.2274	25.7125
23	24.1520	0.8671	22.4526	25.8514
24	23.8542	1.0675	21.7620	25.9465
25	23.5654	1.2428	21.1295	26.0013



Name of Variable = CI	
Mean of Working Series	29.226
Standard Deviation	1.739035
Number of Observations	20



ARIMA Estimation Optimization Summary	
Estimation Method	Conditional Least Squares
Parameters Estimated	3
Termination Criteria	Maximum Relative Change in Estimates
Iteration Stopping Value	0.001
Criteria Value	0.070893
Maximum Absolute Value of Gradient	5.247707
R-Square Change from Last Iteration	0.534727
Objective Function	Sum of Squared Residuals
Objective Function Value	3.22718
Marquardt's Lambda Coefficient	0.00001
Numerical Derivative Perturbation Delta	0.001
Iterations	15
Warning Message	Estimates may not have converged.

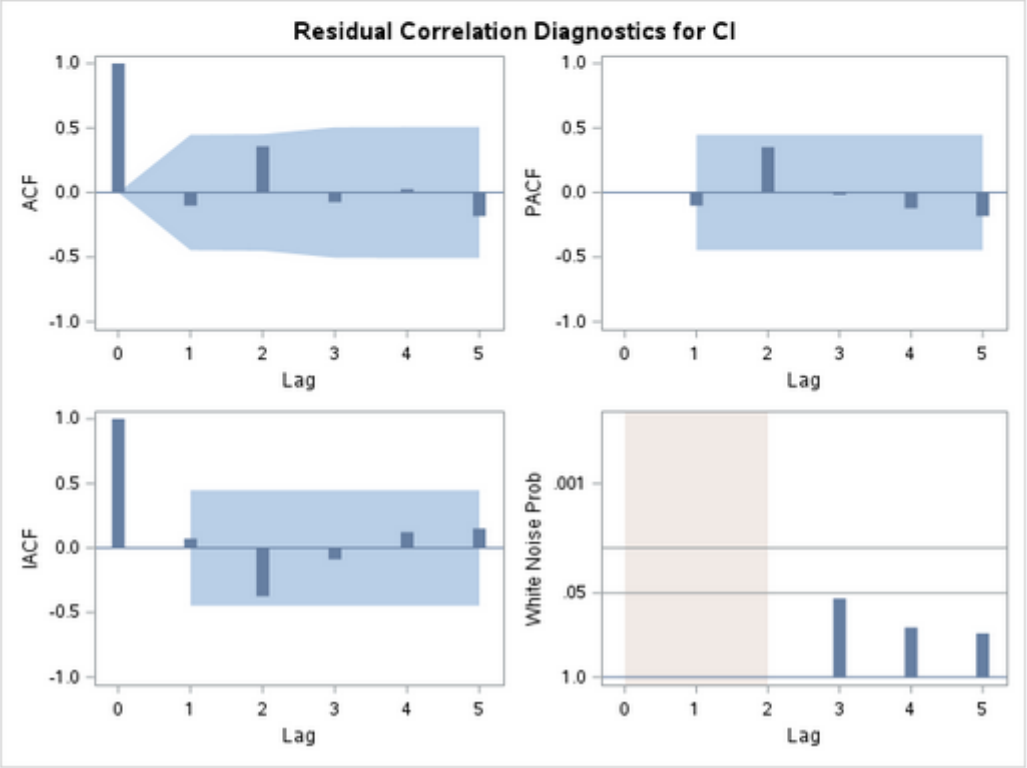
Conditional Least Squares Estimation					
Parameter	Estimate	Standard Error	t Value	Approx Pr > |t|	Lag
MU	30.11611	0.40917	73.60	<.0001	0
AR1,1	1.38761	0.25675	5.40	<.0001	1
AR1,2	-0.38762	0.31097	-1.25	0.2295	2

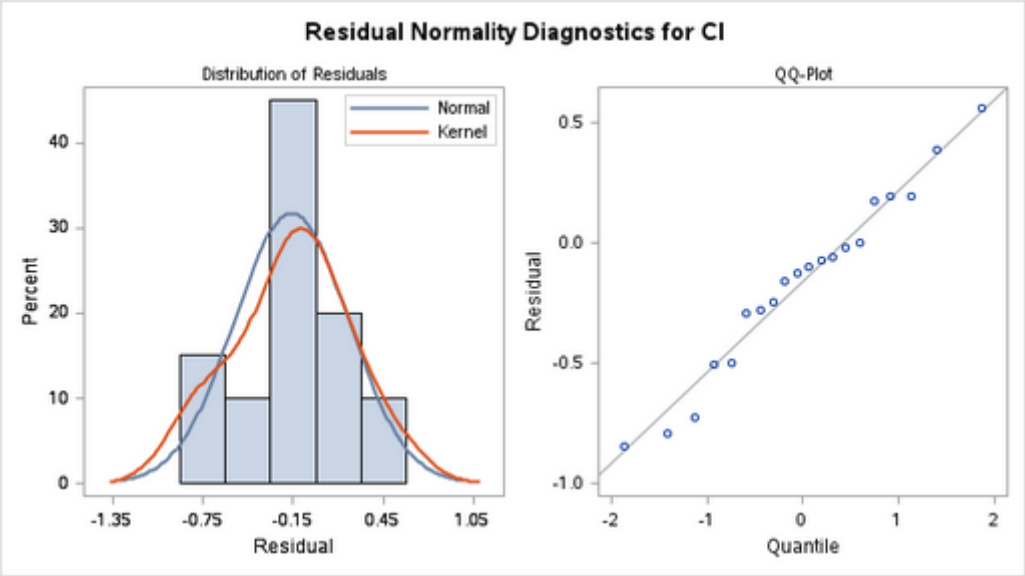
Constant Estimate	0.000047
Variance Estimate	0.189834
Std Error Estimate	0.4357
AIC	26.27507
SBC	29.26227
Number of Residuals	20

* AIC and SBC do not include log determinant.

Correlations of Parameter Estimates			
Parameter	MU	AR1,1	AR1,2
MU	1.000	0.119	-0.116
AR1,1	0.119	1.000	-0.973
AR1,2	-0.116	-0.973	1.000

Autocorrelation Check of Residuals									
To Lag	Chi-Square	DF	Pr > ChiSq	Autocorrelations					
6	6.79	4	0.1472	0.086	0.464	0.079	0.135	-0.032	0.145
12	18.80	10	0.0429	-0.019	0.408	-0.116	0.336	-0.033	0.077
18	23.81	16	0.0938	-0.059	-0.025	-0.047	-0.070	-0.162	0.026

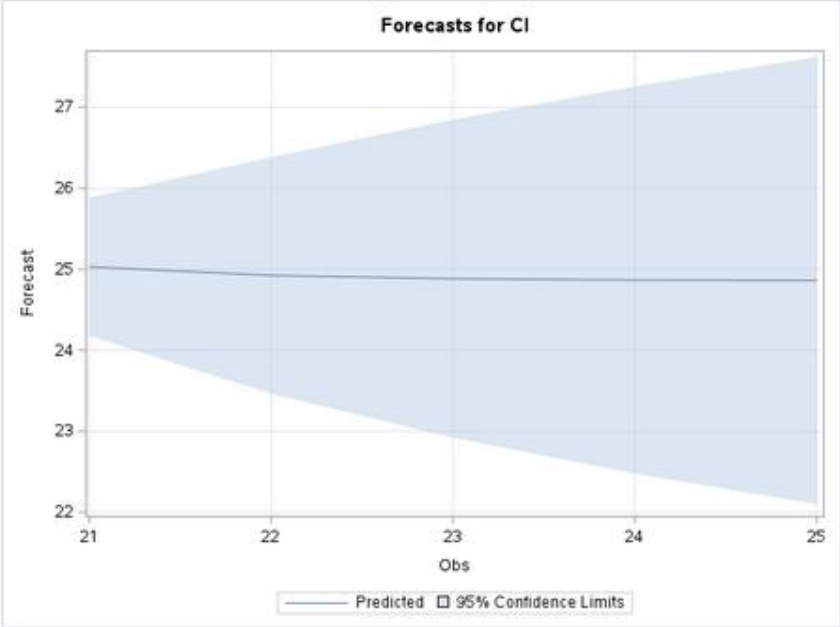




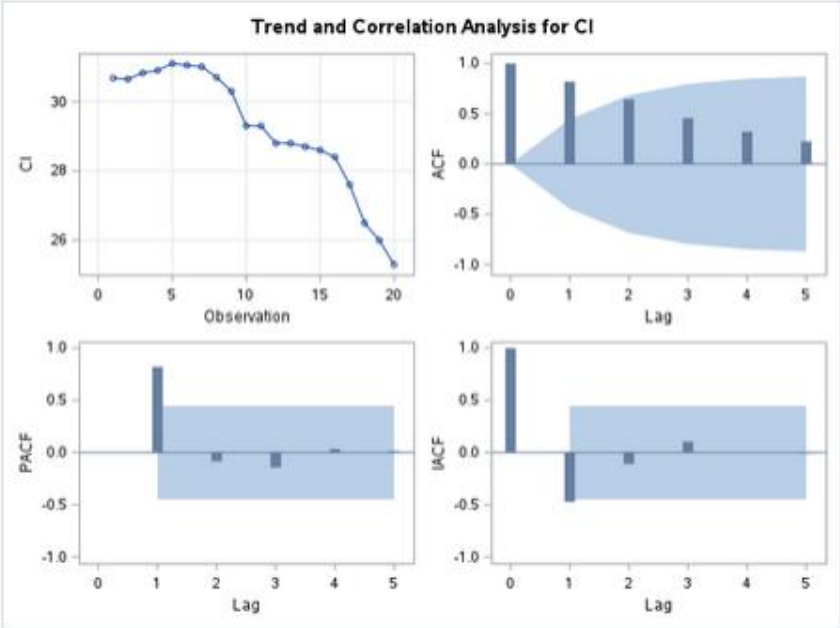
Model for variable CI	
Estimated Mean	30.11611

Autoregressive Factors	
Factor 1:	1 - 1.38761 B**(1) + 0.38762 B**(2)

Forecasts for variable CI				
Obs	Forecast	Std Error	95% Confidence Limits	
21	25.0287	0.4357	24.1747	25.8826
22	24.9235	0.7452	23.4629	26.3841
23	24.8828	1.0022	22.9186	26.8469
24	24.8670	1.2198	22.4762	27.2577
25	24.8609	1.4090	22.0993	27.6225



Name of Variable = CI	
Mean of Working Series	29.226
Standard Deviation	1.739035
Number of Observations	20



ARIMA Estimation Optimization Summary	
Estimation Method	Conditional Least Squares
Parameters Estimated	2
Termination Criteria	Maximum Relative Change in Estimates
Iteration Stopping Value	0.001
Criteria Value	1.07E-14
Maximum Absolute Value of Gradient	0.040116
R-Square Change from Last Iteration	0.00115
Objective Function	Sum of Squared Residuals
Objective Function Value	25.43736
Marquardt's Lambda Coefficient	1E12
Numerical Derivative Perturbation Delta	0.001
Iterations	14
Warning Message	Estimates may not have converged.

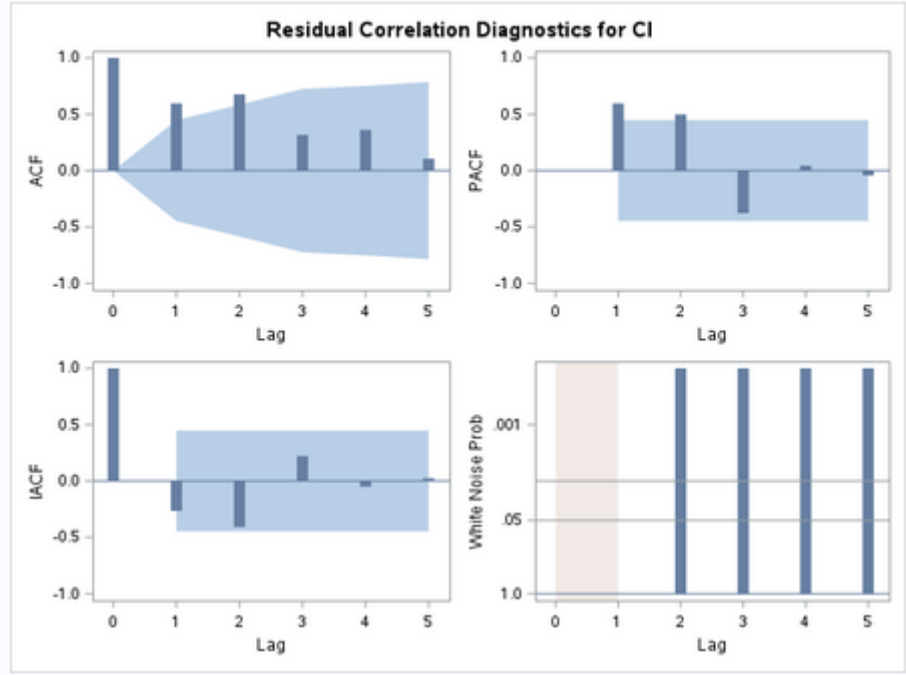
Conditional Least Squares Estimation					
Parameter	Estimate	Standard Error	t Value	Approx Pr > |t|	Lag
MU	29.29857	0.45005	65.10	<.0001	0
MA1,1	-0.78374	0.17293	-4.53	0.0003	1

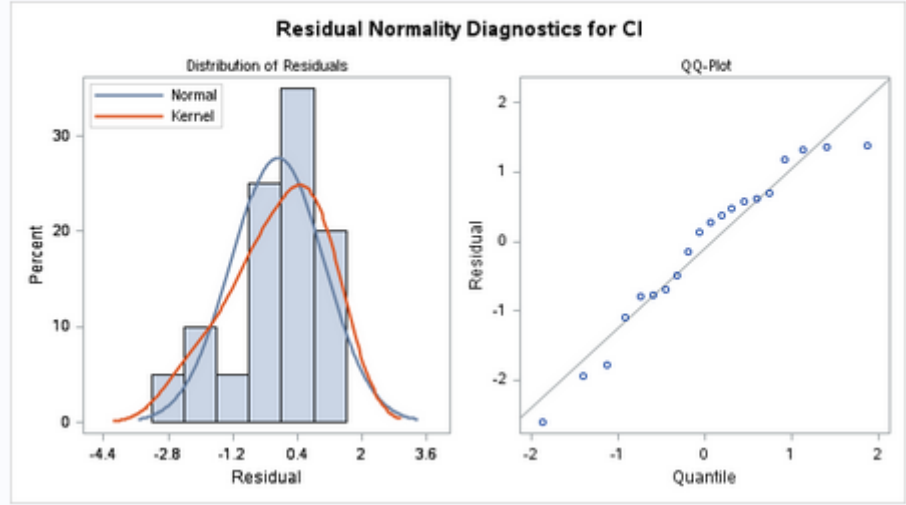
Constant Estimate	29.29857
Variance Estimate	1.413187
Std Error Estimate	1.188775
AIC	65.56728
SBC	67.55874
Number of Residuals	20

* AIC and SBC do not include log determinant.

Correlations of Parameter Estimates		
Parameter	MU	MA1,1
MU	1.000	-0.109
MA1,1	-0.109	1.000

Autocorrelation Check of Residuals									
To Lag	Chi-Square	DF	Pr > ChiSq	Autocorrelations					
6	26.61	5	<.0001	0.595	0.669	0.310	0.353	0.101	0.201
12	38.70	11	<.0001	-0.043	0.063	-0.212	-0.106	-0.340	-0.267
18	82.29	17	<.0001	-0.376	-0.281	-0.334	-0.230	-0.260	-0.124

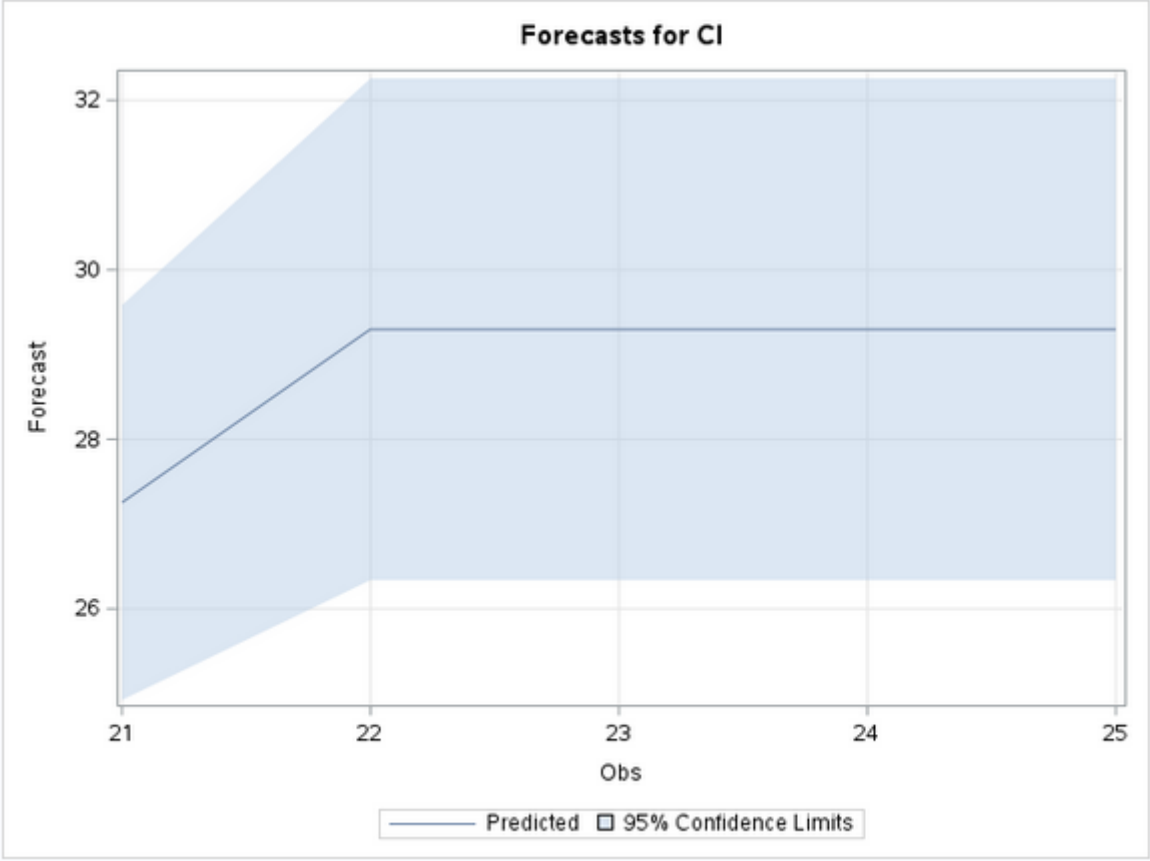




Model for variable CI	
Estimated Mean	29.29857

Moving Average Factors	
Factor 1:	1 + 0.78374 B**(1)

Forecasts for variable CI				
Obs	Forecast	Std Error	95% Confidence Limits	
21	27.2538	1.1888	24.9238	29.5837
22	29.2986	1.5104	26.3383	32.2589
23	29.2986	1.5104	26.3383	32.2589
24	29.2986	1.5104	26.3383	32.2589
25	29.2986	1.5104	26.3383	32.2589



Name of Variable = PI	
Mean of Working Series	1.7605
Standard Deviation	0.462736
Number of Observations	20

Squared Canonical Correlation Estimates						
Lags	MA 0	MA 1	MA 2	MA 3	MA 4	MA 5
AR 0	0.6385	0.1390	0.0132	0.2059	0.4883	0.6205
AR 1	0.5151	0.5103	0.3830	0.3086	0.3424	0.1500
AR 2	0.0996	0.0016	0.2063	0.1600	0.1336	0.0413
AR 3	0.0041	0.0560	0.1652	0.0046	0.0046	0.0115
AR 4	0.3029	0.2599	0.1363	0.0029	0.0089	0.0111
AR 5	0.0124	0.2111	0.2775	0.0160	0.0083	0.0049

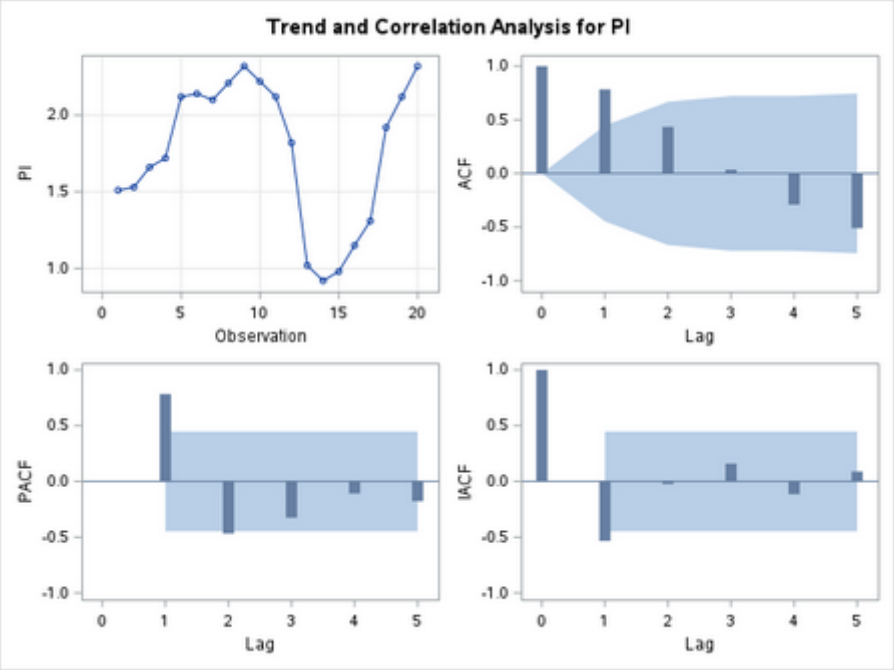
SCAN Chi-Square[1] Probability Values						
Lags	MA 0	MA 1	MA 2	MA 3	MA 4	MA 5
AR 0	<.0001	0.2682	0.7636	0.2429	0.0813	0.0788
AR 1	0.0002	0.0119	0.0360	0.0623	0.0702	0.3162
AR 2	0.1694	0.8862	0.1177	0.1213	0.2428	0.5541
AR 3	0.7905	0.4002	0.1521	0.8311	0.8363	0.7569
AR 4	0.0163	0.0342	0.2683	0.8727	0.7746	0.7680
AR 5	0.6653	0.1089	0.0672	0.6870	0.7857	0.8565

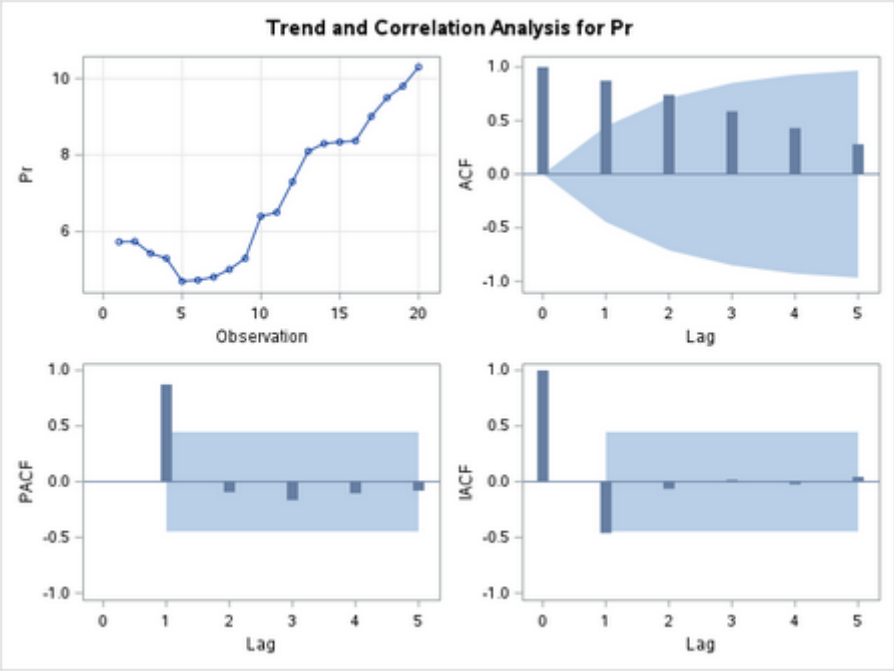
Extended Sample Autocorrelation Function						
Lags	MA 0	MA 1	MA 2	MA 3	MA 4	MA 5
AR 0	0.7836	0.4336	0.0341	-0.2909	-0.5101	-0.5481
AR 1	0.6527	0.4474	0.1089	-0.1483	-0.4331	-0.2617
AR 2	-0.4942	-0.0342	0.0627	0.1710	-0.4560	-0.3847
AR 3	-0.3939	-0.1032	0.1135	-0.2603	-0.4759	0.0532
AR 4	-0.3981	-0.2705	0.1631	-0.2510	-0.4104	-0.4888
AR 5	0.6375	0.2874	0.0023	-0.3856	-0.4958	.

ESACF Probability Values						
Lags	MA 0	MA 1	MA 2	MA 3	MA 4	MA 5
AR 0	0.0005	0.1939	0.9247	0.4204	0.1710	0.1770
AR 1	0.0044	0.1971	0.7647	0.6394	0.1337	0.4190
AR 2	0.0360	0.8998	0.8226	0.5292	0.2109	0.3466
AR 3	0.1044	0.7027	0.6846	0.4673	0.1867	0.8656
AR 4	0.1113	0.3143	0.5780	0.4732	0.2697	0.2787
AR 5	0.0135	0.4266	0.9948	0.2747	0.2538	.

ARMA(p+d,q) Tentative Order Selection Tests			
SCAN		ESACF	
p+d	q	p+d	q
0	3	0	1
2	2	1	1
5	0	3	0
		2	1
		4	0

(5% Significance Level)





The ARIMA Procedure

Name of Variable = Pr	
Mean of Working Series	6.9335
Standard Deviation	1.828232
Number of Observations	20

Squared Canonical Correlation Estimates						
Lags	MA 0	MA 1	MA 2	MA 3	MA 4	MA 5
AR 0	0.9267	0.7736	0.5636	0.3429	0.1577	0.0258
AR 1	0.2606	0.1776	0.1489	0.2082	0.2672	0.2321
AR 2	0.0379	0.0015	0.0073	0.0035	0.0063	0.0178
AR 3	0.0138	0.0035	0.0034	0.0009	0.0005	0.0021
AR 4	0.0322	0.0046	0.0010	0.0009	0.0020	0.0001
AR 5	0.0148	0.0081	0.0006	<.0001	0.0006	0.0033

SCAN Chi-Square[1] Probability Values						
Lags	MA 0	MA 1	MA 2	MA 3	MA 4	MA 5
AR 0	<.0001	0.0084	0.0769	0.2132	0.4102	0.7374
AR 1	0.0166	0.0940	0.1925	0.1709	0.2757	0.2773
AR 2	0.4044	0.8951	0.7750	0.8476	0.8097	0.6988
AR 3	0.6271	0.8543	0.8628	0.9255	0.9436	0.8886
AR 4	0.4692	0.8387	0.9231	0.9294	0.9050	0.9745
AR 5	0.6366	0.7941	0.9393	0.9920	0.9476	0.8799

Extended Sample Autocorrelation Function						
Lags	MA 0	MA 1	MA 2	MA 3	MA 4	MA 5
AR 0	0.8735	0.7405	0.5865	0.4304	0.2793	0.1220
AR 1	0.2849	0.2682	0.1702	0.0424	-0.2641	-0.3196
AR 2	0.5149	0.1134	0.1927	-0.0037	-0.3631	-0.0264
AR 3	-0.0341	0.5003	-0.0214	-0.0250	-0.4305	-0.0503
AR 4	0.7227	0.3811	-0.2122	-0.1410	-0.3596	-0.0040
AR 5	0.5607	0.3102	-0.0921	-0.1916	-0.4619	.

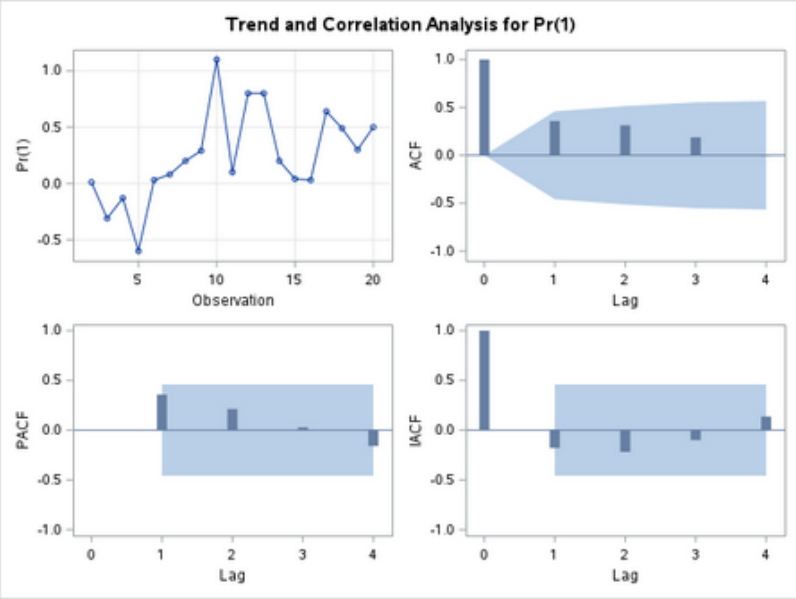
ESACF Probability Values						
Lags	MA 0	MA 1	MA 2	MA 3	MA 4	MA 5
AR 0	<.0001	0.0372	0.1682	0.3539	0.5637	0.8041
AR 1	0.2143	0.2912	0.5400	0.8802	0.3206	0.2534
AR 2	0.0289	0.6507	0.5412	0.9893	0.2788	0.9201
AR 3	0.8883	0.0650	0.9469	0.9183	0.1979	0.8688
AR 4	0.0038	0.1339	0.3990	0.6333	0.3454	0.9933
AR 5	0.0299	0.2344	0.7499	0.6051	0.2405	.

ARMA(p+d,q) Tentative Order Selection Tests			
SCAN		ESACF	
p+d	q	p+d	q
1	1	1	0
2	0	3	0
0	2	0	2
		5	1

(5% Significance Level)

The ARIMA Procedure

Name of Variable = Pr	
Period(s) of Differencing	1
Mean of Working Series	0.240526
Standard Deviation	0.399835
Number of Observations	19
Observation(s) eliminated by differencing	1



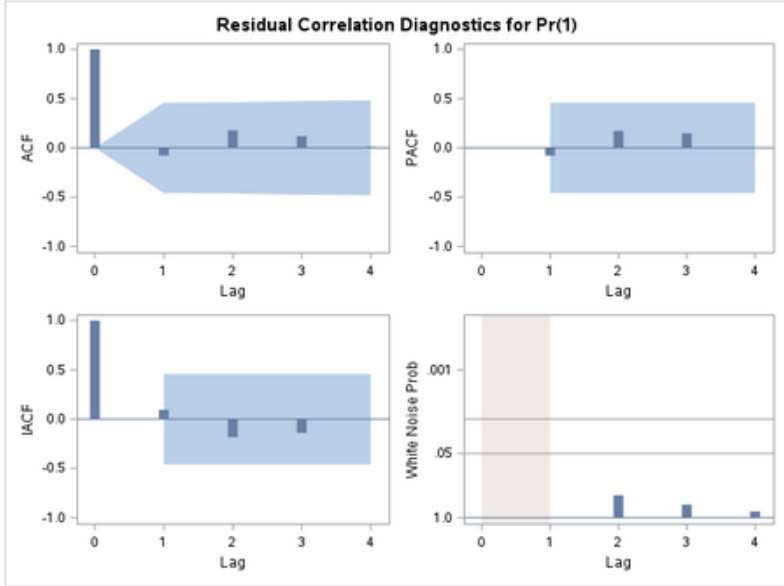
Conditional Least Squares Estimation					
Parameter	Estimate	Standard Error	t Value	Approx Pr > |t|	Lag
MU	0.23759	0.13739	1.73	0.1019	0
AR1,1	0.36623	0.22877	1.60	0.1278	1

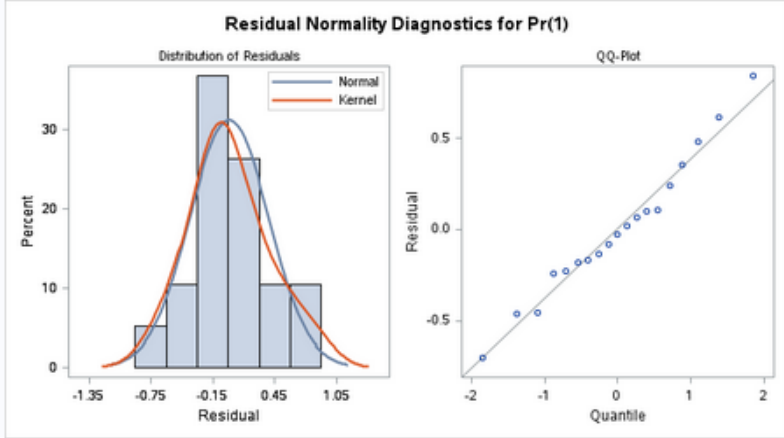
Constant Estimate	0.150575
Variance Estimate	0.155262
Std Error Estimate	0.394033
AIC	20.41622
SBC	22.3051
Number of Residuals	19

* AIC and SBC do not include log determinant.

Correlations of Parameter Estimates		
Parameter	MU	AR1,1
MU	1.000	0.027
AR1,1	0.027	1.000

Autocorrelation Check of Residuals									
To Lag	Chi-Square	DF	Pr > ChiSq	Autocorrelations					
6	2.40	5	0.7918	-0.078	0.178	0.120	0.013	-0.189	-0.068
12	3.57	11	0.9808	-0.112	-0.025	-0.074	0.006	0.080	-0.059
18	5.94	17	0.9936	-0.027	-0.062	-0.122	-0.019	-0.040	-0.021

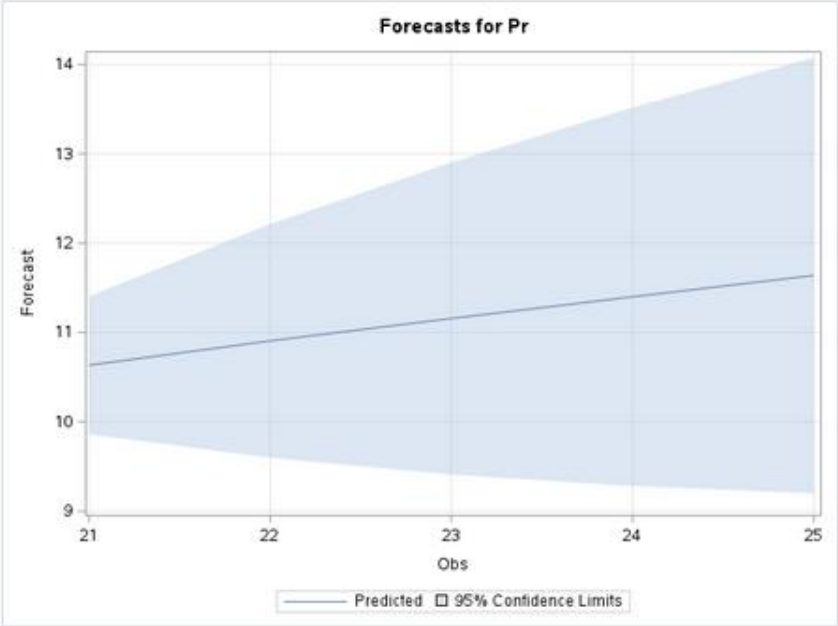




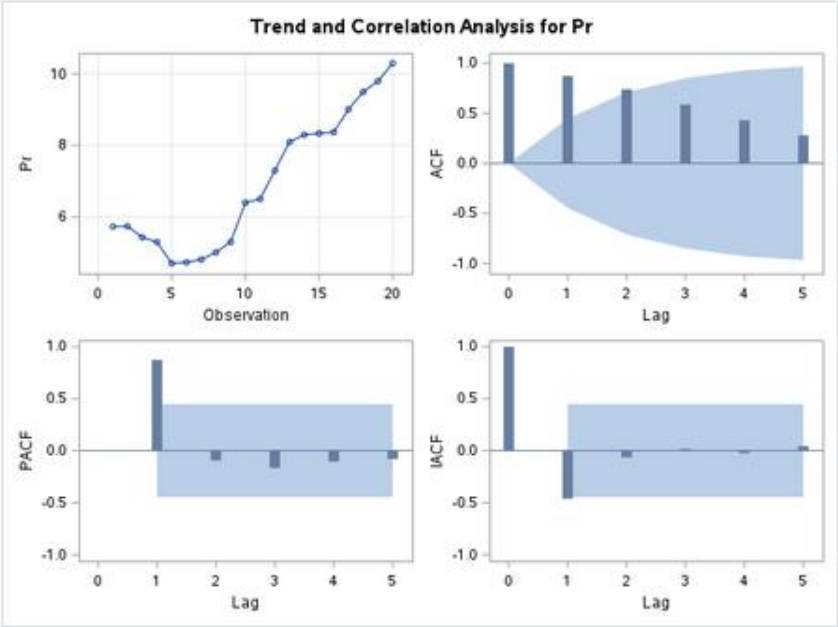
Model for variable Pr	
Estimated Mean	0.237588
Period(s) of Differencing	1

Autoregressive Factors	
Factor 1:	1 - 0.36623 B**(1)

Forecasts for variable Pr				
Obs	Forecast	Std Error	95% Confidence Limits	
21	10.6337	0.3940	9.8614	11.4060
22	10.9065	0.6671	9.5989	12.2140
23	11.1570	0.8914	9.4099	12.9041
24	11.3993	1.0804	9.2816	13.5169
25	11.6386	1.2445	9.1994	14.0778



Name of Variable = Pr	
Mean of Working Series	6.9335
Standard Deviation	1.828232
Number of Observations	20



ARIMA Estimation Optimization Summary	
Estimation Method	Conditional Least Squares
Parameters Estimated	2
Termination Criteria	Maximum Relative Change in Estimates
Iteration Stopping Value	0.001
Criteria Value	0.101796
Maximum Absolute Value of Gradient	5.406361
R-Square Change from Last Iteration	0.451442
Objective Function	Sum of Squared Residuals
Objective Function Value	4.558432
Marquardt's Lambda Coefficient	0.00001
Numerical Derivative Perturbation Delta	0.001
Iterations	14
Warning Message	Estimates may not have converged.

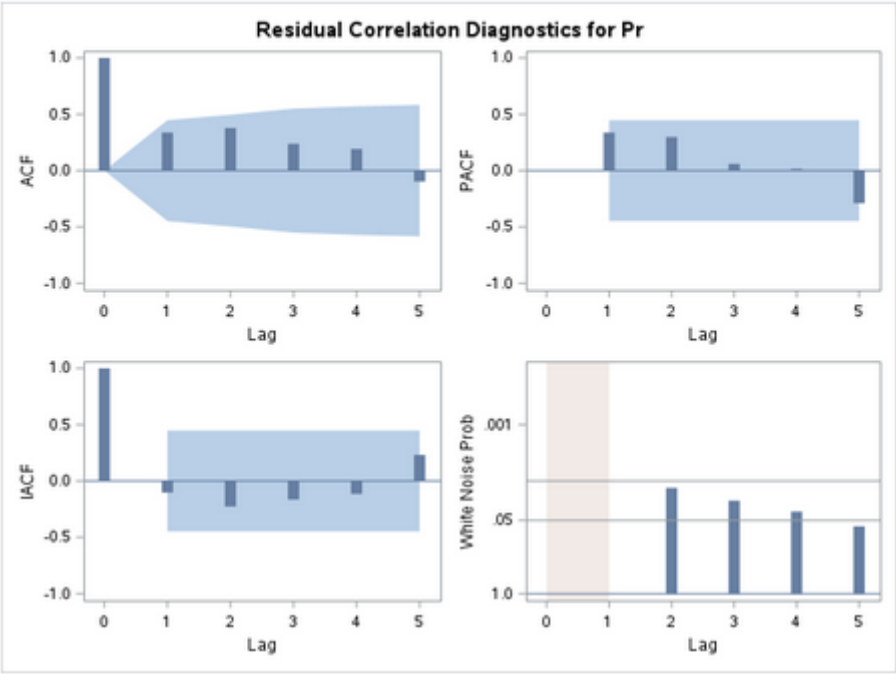
Conditional Least Squares Estimation					
Parameter	Estimate	Standard Error	t Value	Approx Pr > |t|	Lag
MU	6.37940	0.50324	12.68	<.0001	0
AR1,1	1.00000	0.06630	15.08	<.0001	1

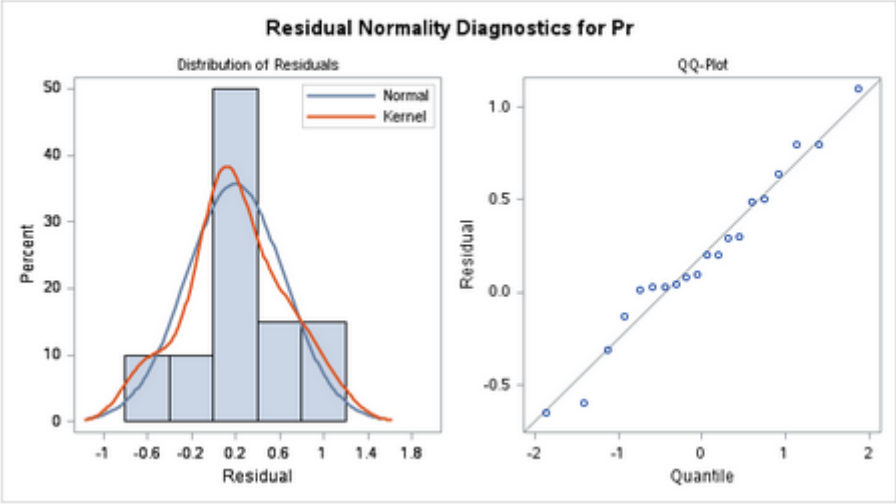
Constant Estimate	3.962E-6
Variance Estimate	0.253246
Std Error Estimate	0.503236
AIC	31.18247
SBC	33.17393
Number of Residuals	20

* AIC and SBC do not include log determinant.

Correlations of Parameter Estimates		
Parameter	MU	AR1,1
MU	1.000	-0.000
AR1,1	-0.000	1.000

Autocorrelation Check of Residuals									
To Lag	Chi-Square	DF	Pr > ChiSq	Autocorrelations					
6	18.53	5	0.0024	0.464	0.493	0.377	0.325	0.118	0.137
12	21.97	11	0.0246	0.107	0.083	-0.117	0.095	-0.073	-0.171
18	28.09	17	0.0439	-0.102	-0.099	-0.111	-0.125	-0.103	-0.042

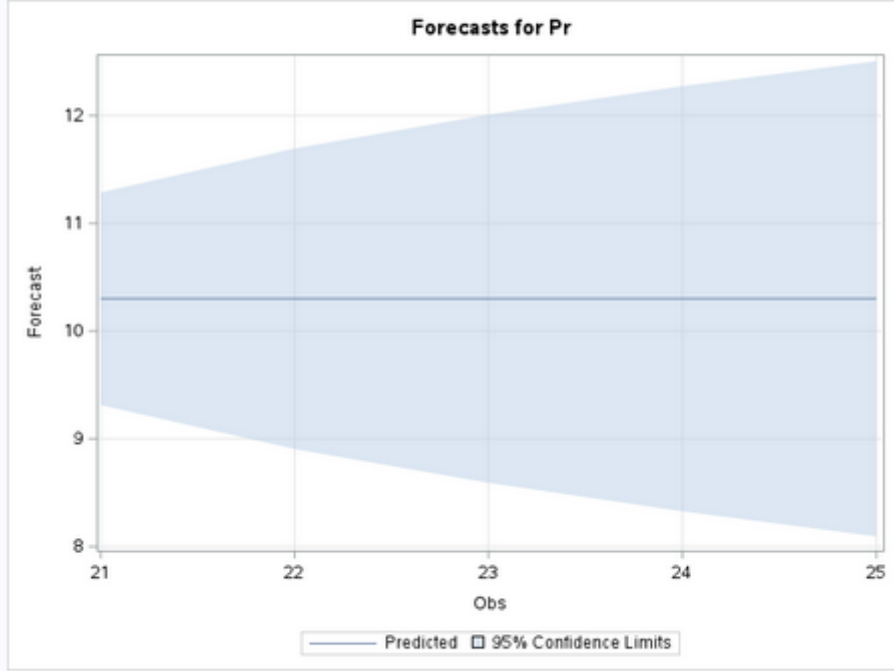




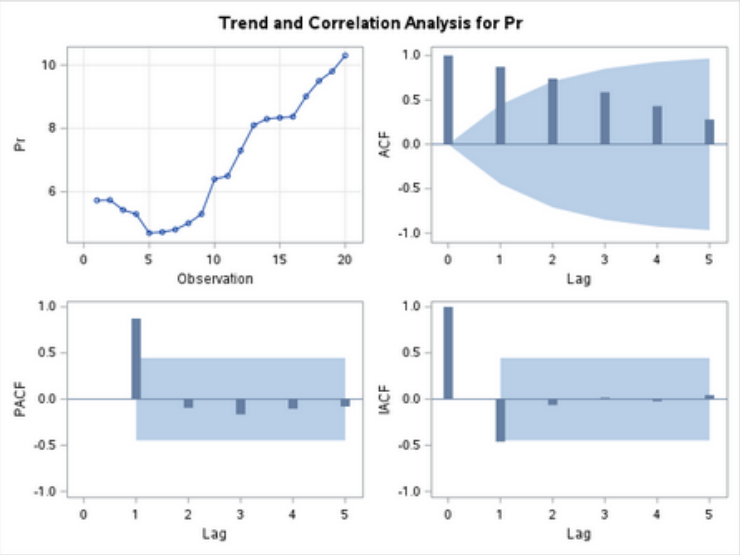
Model for variable Pr	
Estimated Mean	6.379404

Autoregressive Factors	
Factor 1:	1 - 1 B**(1)

Forecasts for variable Pr				
Obs	Forecast	Std Error	95% Confidence Limits	
21	10.3000	0.5032	9.3137	11.2863
22	10.3000	0.7117	8.9051	11.6949
23	10.3000	0.8716	8.5916	12.0084
24	10.3000	1.0065	8.3273	12.2726
25	10.3000	1.1253	8.0945	12.5055



Name of Variable = Pr	
Mean of Working Series	6.9335
Standard Deviation	1.828232
Number of Observations	20



ARIMA Estimation Optimization Summary	
Estimation Method	Conditional Least Squares
Parameters Estimated	3
Termination Criteria	Maximum Relative Change in Estimates
Iteration Stopping Value	0.001
Criteria Value	0.320752
Maximum Absolute Value of Gradient	3.082666
R-Square Change from Last Iteration	0.367402
Objective Function	Sum of Squared Residuals
Objective Function Value	10.40098
Marquardt's Lambda Coefficient	0.00001
Numerical Derivative Perturbation Delta	0.001
Iterations	19
Warning Message	Estimates may not have converged.

Conditional Least Squares Estimation					
Parameter	Estimate	Standard Error	t Value	Approx Pr > |t|	Lag
MU	6.73081	0.43755	15.38	<.0001	0
MA1,1	-1.12926	0.16102	-7.01	<.0001	1
MA1,2	-1.00000	0.21588	-4.63	0.0002	2

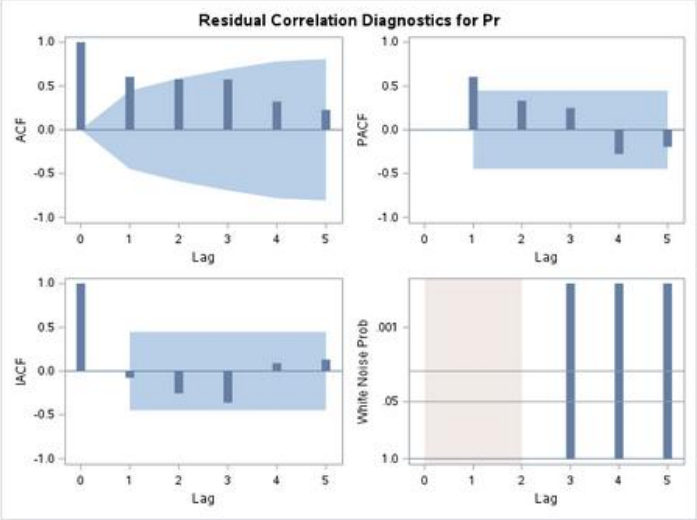
Constant Estimate	6.730812
Variance Estimate	0.611822
Std Error Estimate	0.782191
AIC	49.68089
SBC	52.66809
Number of Residuals	20

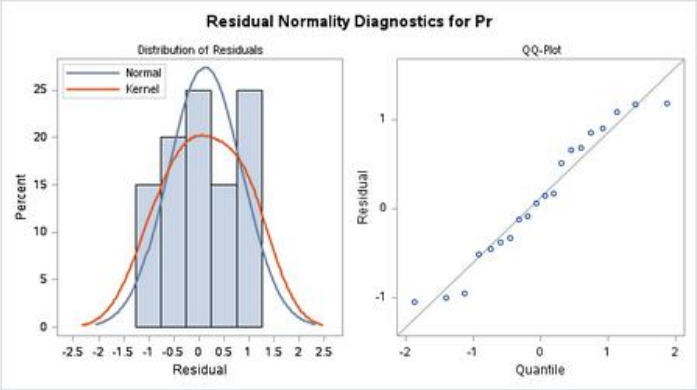
* AIC and SBC do not include log determinant.

* AIC and SBC do not include log determinant.

Correlations of Parameter Estimates			
Parameter	MU	MA1,1	MA1,2
MU	1.000	0.436	0.726
MA1,1	0.436	1.000	0.664
MA1,2	0.726	0.664	1.000

Autocorrelation Check of Residuals									
To Lag	Chi-Square	DF	Pr > ChiSq	Autocorrelations					
6	30.12	4	<.0001	0.615	0.575	0.566	0.327	0.231	0.126
12	48.97	10	<.0001	0.061	-0.073	-0.280	-0.187	-0.327	-0.396
18	78.29	16	<.0001	-0.308	-0.297	-0.277	-0.217	-0.149	-0.088

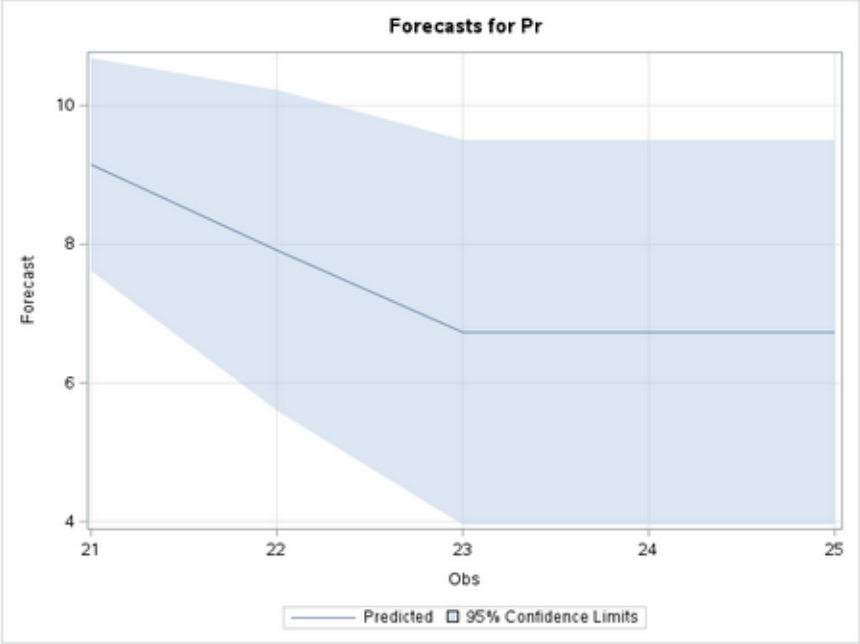




Model for variable Pr	
Estimated Mean	6.730812

Moving Average Factors	
Factor 1:	1 + 1.12926 B**(1) + 1 B**(2)

Forecasts for variable Pr				
Obs	Forecast	Std Error	95% Confidence Limits	
21	9.1503	0.7822	7.6172	10.6834
22	7.9141	1.1798	5.6017	10.2266
23	6.7308	1.4156	3.9563	9.5053
24	6.7308	1.4156	3.9563	9.5053
25	6.7308	1.4156	3.9563	9.5053



The ARIMA Procedure

Name of Variable = PI	
Mean of Working Series	1.7605
Standard Deviation	0.462736
Number of Observations	20

Squared Canonical Correlation Estimates						
Lags	MA 0	MA 1	MA 2	MA 3	MA 4	MA 5
AR 0	0.6385	0.1390	0.0132	0.2059	0.4883	0.6205
AR 1	0.5151	0.5103	0.3830	0.3086	0.3424	0.1500
AR 2	0.0996	0.0016	0.2063	0.1600	0.1336	0.0413
AR 3	0.0041	0.0560	0.1652	0.0046	0.0046	0.0115
AR 4	0.3029	0.2599	0.1363	0.0029	0.0089	0.0111
AR 5	0.0124	0.2111	0.2775	0.0160	0.0083	0.0049

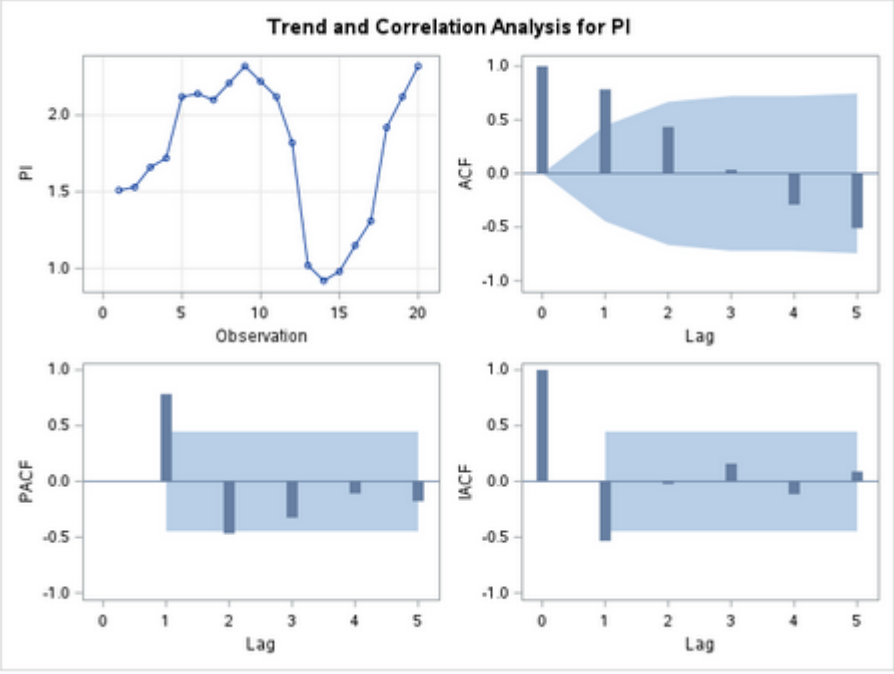
SCAN Chi-Square[1] Probability Values						
Lags	MA 0	MA 1	MA 2	MA 3	MA 4	MA 5
AR 0	<.0001	0.2682	0.7636	0.2429	0.0813	0.0788
AR 1	0.0002	0.0119	0.0360	0.0623	0.0702	0.3162
AR 2	0.1694	0.8862	0.1177	0.1213	0.2428	0.5541
AR 3	0.7905	0.4002	0.1521	0.8311	0.8363	0.7569
AR 4	0.0163	0.0342	0.2683	0.8727	0.7746	0.7680
AR 5	0.6653	0.1089	0.0672	0.6870	0.7857	0.8565

Extended Sample Autocorrelation Function						
Lags	MA 0	MA 1	MA 2	MA 3	MA 4	MA 5
AR 0	0.7836	0.4336	0.0341	-0.2909	-0.5101	-0.5481
AR 1	0.6527	0.4474	0.1089	-0.1483	-0.4331	-0.2617
AR 2	-0.4942	-0.0342	0.0627	0.1710	-0.4560	-0.3847
AR 3	-0.3939	-0.1032	0.1135	-0.2603	-0.4759	0.0532
AR 4	-0.3981	-0.2705	0.1631	-0.2510	-0.4104	-0.4888
AR 5	0.6375	0.2874	0.0023	-0.3856	-0.4958	.

ESACF Probability Values						
Lags	MA 0	MA 1	MA 2	MA 3	MA 4	MA 5
AR 0	0.0005	0.1939	0.9247	0.4204	0.1710	0.1770
AR 1	0.0044	0.1971	0.7647	0.6394	0.1337	0.4190
AR 2	0.0360	0.8998	0.8226	0.5292	0.2109	0.3466
AR 3	0.1044	0.7027	0.6846	0.4673	0.1867	0.8656
AR 4	0.1113	0.3143	0.5780	0.4732	0.2697	0.2787
AR 5	0.0135	0.4266	0.9948	0.2747	0.2538	.

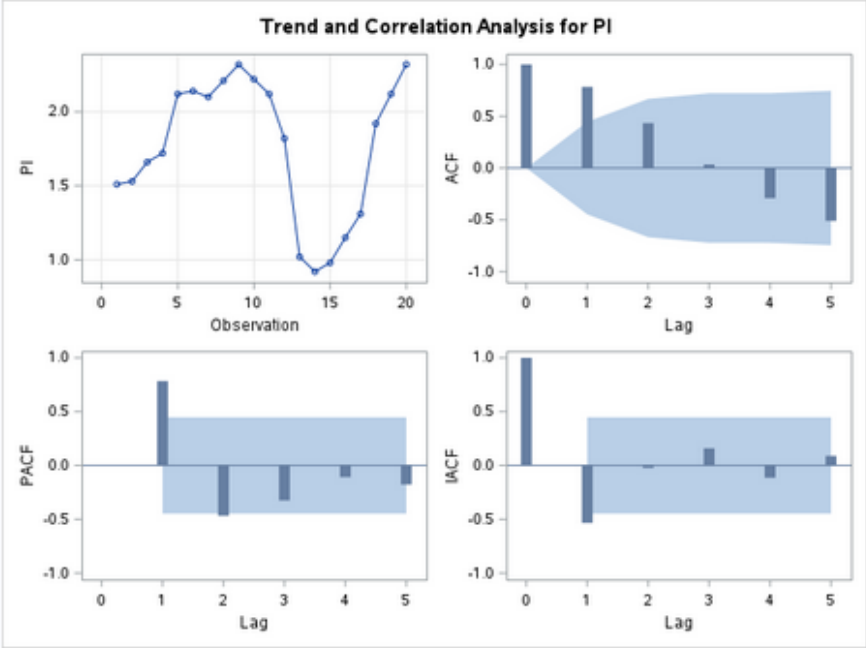
ARMA(p+d,q) Tentative Order Selection Tests			
SCAN		ESACF	
p+d	q	p+d	q
0	3	0	1
2	2	1	1
5	0	3	0
		2	1
		4	0

(5% Significance Level)



The ARIMA Procedure

Name of Variable = PI	
Mean of Working Series	1.7605
Standard Deviation	0.462736
Number of Observations	20



ARIMA Estimation Optimization Summary	
Estimation Method	Conditional Least Squares
Parameters Estimated	4
Termination Criteria	Maximum Relative Change in Estimates
Iteration Stopping Value	0.001
Criteria Value	0.475442
Maximum Absolute Value of Gradient	0.329222
R-Square Change from Last Iteration	0.287822
Objective Function	Sum of Squared Residuals
Objective Function Value	0.978784
Marquardt's Lambda Coefficient	0.00001
Numerical Derivative Perturbation Delta	0.001
Iterations	16
Warning Message	Estimates may not have converged.

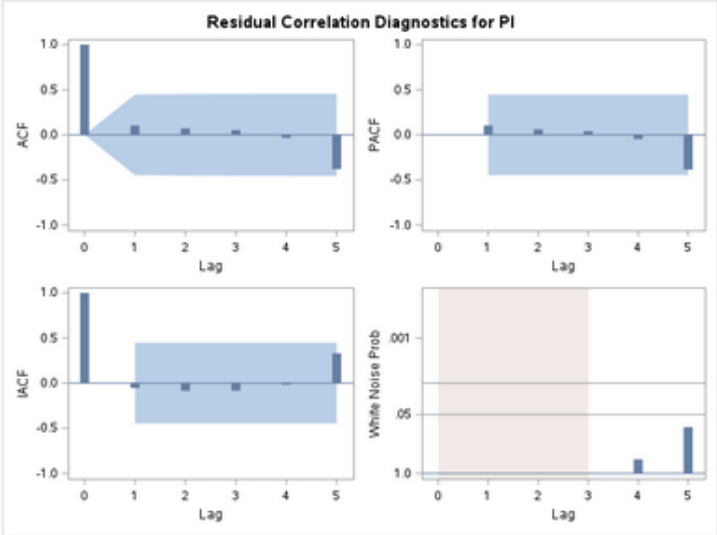
Conditional Least Squares Estimation					
Parameter	Estimate	Standard Error	t Value	Approx Pr > |t|	Lag
MU	1.72510	0.12569	13.73	<.0001	0
MA1,1	-1.10469	0.28524	-3.87	0.0013	1
MA1,2	-1.01996	0.33165	-3.08	0.0072	2
MA1,3	-0.91527	0.46860	-1.95	0.0685	3

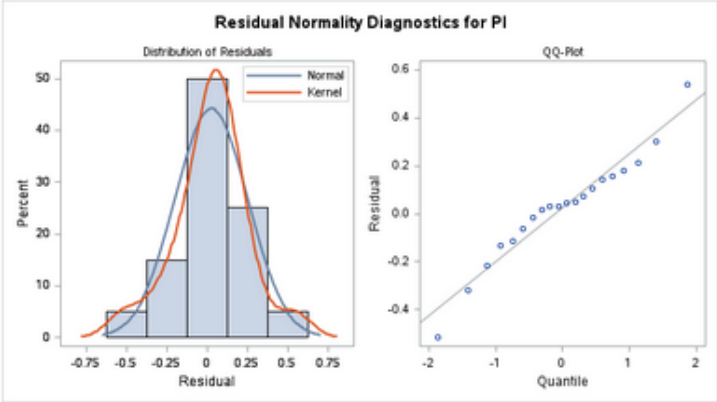
Constant Estimate	1.725105
Variance Estimate	0.061174
Std Error Estimate	0.247334
AIC	4.414002
SBC	8.398931
Number of Residuals	20

* AIC and SBC do not include log determinant.

Correlations of Parameter Estimates				
Parameter	MU	MA1,1	MA1,2	MA1,3
MU	1.000	0.230	0.234	0.134
MA1,1	0.230	1.000	0.782	0.675
MA1,2	0.234	0.782	1.000	0.753
MA1,3	0.134	0.675	0.753	1.000

Autocorrelation Check of Residuals										
To Lag	Chi-Square	DF	Pr > ChiSq	Autocorrelations						
6	6.03	3	0.1103	0.118	0.080	0.047	-0.031	-0.377	-0.203	
12	12.79	9	0.1722	-0.177	-0.348	0.088	0.029	0.094	0.109	
18	17.88	15	0.2889	0.153	0.014	0.093	0.115	-0.093	-0.023	

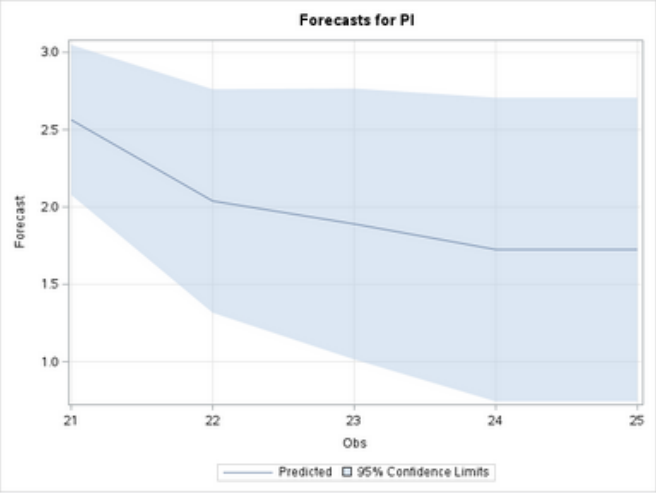




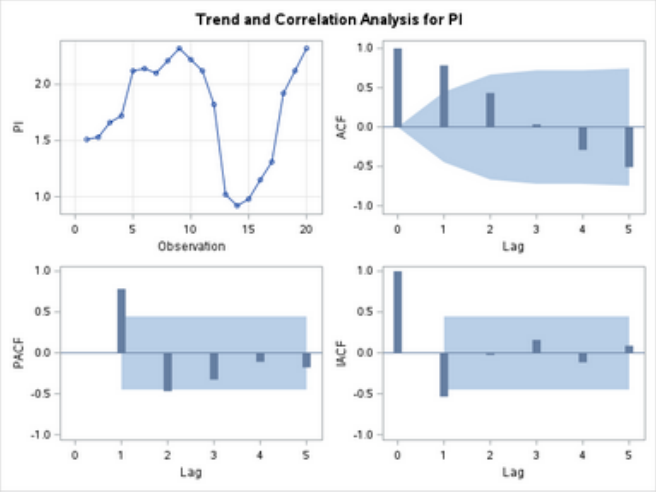
Model for variable PI	
Estimated Mean	1.725105

Moving Average Factors	
Factor 1:	1 + 1.10469 B**(1) + 1.01996 B**(2) + 0.91527 B**(3)

Forecasts for variable PI				
Obs	Forecast	Std Error	95% Confidence Limits	
21	2.5834	0.2473	2.0788	3.0481
22	2.0387	0.3885	1.3184	2.7611
23	1.8905	0.4488	1.0151	2.7658
24	1.7251	0.5007	0.7437	2.7065
25	1.7251	0.5007	0.7437	2.7065



Name of Variable = PI	
Mean of Working Series	1.7605
Standard Deviation	0.462736
Number of Observations	20



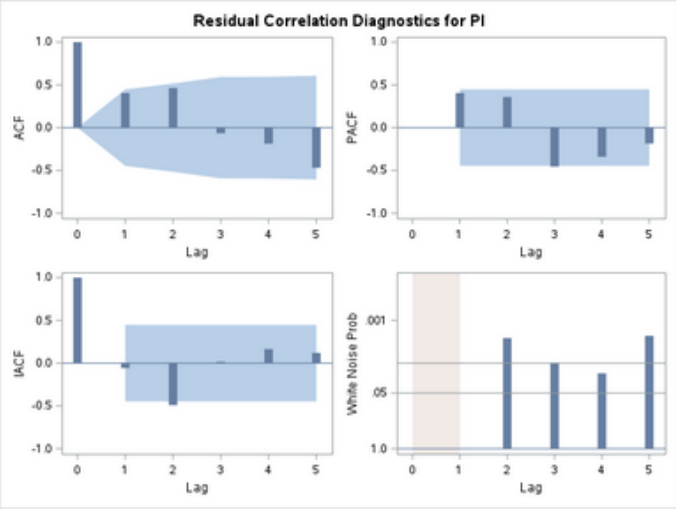
Conditional Least Squares Estimation					
Parameter	Estimate	Standard Error	t Value	Approx. Pr > |t|	Lag
MU	1.76852	0.12388	14.30	<.0001	0
MA1,1	-0.83194	0.21136	-3.94	0.0010	1

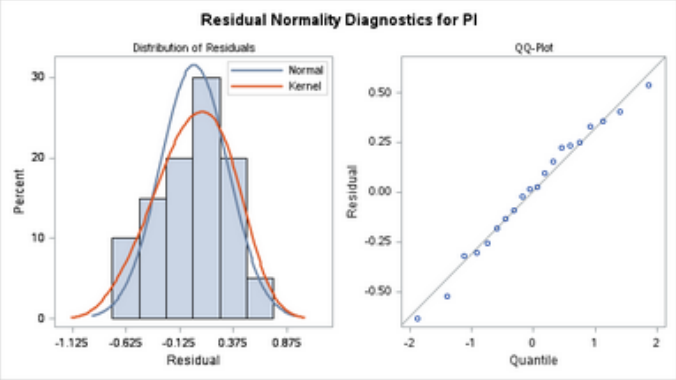
Constant Estimate	1.768515
Variance Estimate	0.105342
Std Error Estimate	0.324585
AIC	13.63952
SBC	15.63099
Number of Residuals	20

* AIC and SBC do not include log determinant.

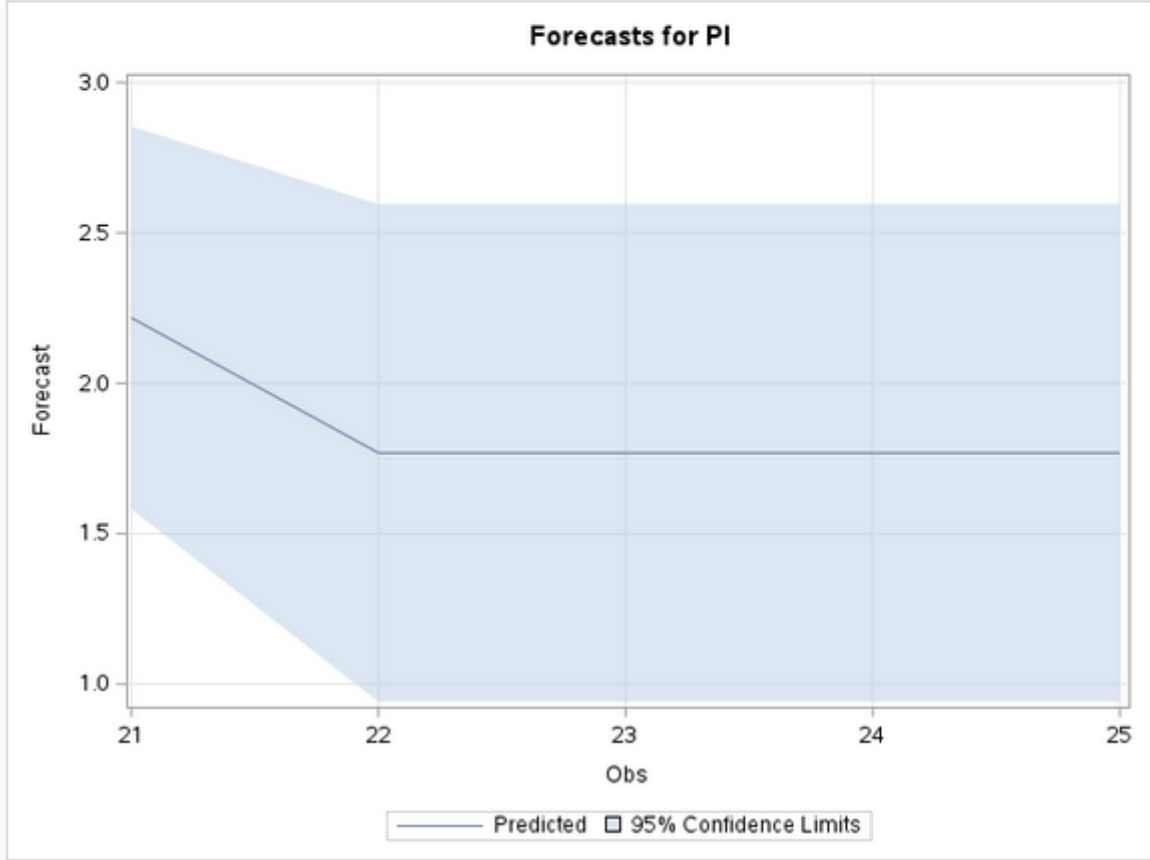
Correlations of Parameter Estimates		
Parameter	MU	MA1,1
MU	1.000	-0.021
MA1,1	-0.021	1.000

Autocorrelation Check of Residuals									
To Lag	Chi-Square	DF	Pr > ChiSq	Autocorrelations					
6	22.09	5	0.0005	0.405	0.483	-0.067	-0.188	-0.470	-0.417
12	34.09	11	0.0003	-0.398	-0.343	-0.029	-0.048	0.144	0.148
18	39.79	17	0.0014	0.197	0.123	0.104	0.043	-0.081	-0.009





Model for variable PI	
Estimated Mean	1.768515

Moving Average Factors	
Factor 1:	1 + 0.83194 B**(1)

Forecasts for variable PI				
Obs	Forecast	Std Error	95% Confidence Limits	
21	2.2179	0.3246	1.5818	2.8540
22	1.7685	0.4222	0.9410	2.5960
23	1.7685	0.4222	0.9410	2.5960
24	1.7685	0.4222	0.9410	2.5960
25	1.7685	0.4222	0.9410	2.5960